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Was the Lhasa Block at Low Latitudes in the Middle Permian? Insights From New Paleomagnetic and Geochronological Data

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Abstract The absence of reliable paleomagnetic constraints from the Lhasa Block has led to alternative interpretations of its late Paleozoic position and timing of rifting from Gondwana, reflecting uncertainties in early Neo-Tethyan paleogeography. This study presents paleomagnetic and geochronological data from the middle Permian Luobadui Formation, providing a new paleogeographic constraint on the Lhasa Block. Despite possible remagnetization, the dual-polarity magnetization, hosted in different minerals and lithologies, likely represents a middle Permian remanence. This constraint implies the Lhasa Block was located at 16.7 ± 5.3°S at 267.8 ± 5 Ma, following its rifting from Gondwana. New U-Pb detrital zircon ages from sandstones further suggest the Lhasa Block was located along the northwestern margin of Australia prior to rifting. Integrating other geological evidence, we propose that the Bangong Co-Nujiang and Yarlung-Zangbo oceans, now preserved as sutures flanking the Lhasa Block, both opened before the middle Permian, potentially representing branches of the same nascent oceanic corridor (Neo-Tethys).

Plain Language Summary The Lhasa Block, located in southern Tibet, is critical to understanding the early tectonic evolution of the Neo-Tethys Ocean, which separated Gondwana from Laurasia during the late Paleozoic to early Mesozoic. However, the timing and mechanisms of the Lhasa Block's rifting from Gondwana remain controversial, with several competing models proposing different scenarios. In this study, we present new paleomagnetic and geochronological data from middle Permian strata of the Lhasa Block, which suggest that this block may have rifted from the northwestern margin of the Australian sector of Gondwana before ~268 Ma. Combined with other geological evidence, we propose that the Bangong Co-Nujiang and Yarlung-Zangbo oceans, now preserved as sutures to the north and south of the present Lhasa Block, were branches of the Neo-Tethys Ocean and probably existed side by side in an east-west orientation during the middle Permian. This study thus provides important insights into the tectonic history of the Lhasa Block and the early evolution of the Neo-Tethys Ocean.

Key Points:

- The Lhasa Block was positioned at approximately 17°S around 268 Ma
- The Lhasa Block may have rifted from the northwestern margin of the Australian sector of Gondwana before the middle Permian
- The Bangong Co-Nujiang and Yarlung-Zangbo oceans likely opened before the middle Permian, possibly as branches of the nascent Neo-Tethys

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

The conventional model of the Neo-Tethys Ocean involves its late Paleozoic opening by the separation of a single, elongate continental ribbon—the "Cimmerian continent"—from the northern margin of Gondwana (Stampfli & Borel, 2002). However, in the east there are multiple sutures crossing the southern Tibetan Plateau that have been proposed as correlatives of the Neo-Tethys in the west (Hu et al., 2022), and it remains unclear how these different oceanic corridors developed and relate to one another. The Lhasa Block is particularly important to unraveling this history, as it is bounded to the north by the Bangong Co-Nujiang Suture Zone (BNS) and to the south by the Yarlung-Zangbo Suture Zone (YZS)—both of which have been proposed to mark the relics of former eastwards continuations of the Neo-Tethys Ocean.

If the YZS represents a former continuation of the Neo-Tethys Ocean, the Lhasa Block could have split away from Gondwana with the rest of the Cimmerian continent, perhaps not unlike the scenario envisioned in the conventional model. However, many alternative models consider the Yarlung-Zangbo Ocean (YZO) to be distinctly younger—opening in the Late Triassic (Meng et al., 2023). In this case, the Lhasa Block would have spalled from

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northern Gondwana independently of the Cimmerian continent, and possibly as an isolated continental block. In addition to these temporal uncertainties, the pre-rift location of the Lhasa Block along the northern margin of Gondwana is also disputed. Conventionally, the Lhasa Block has been reconstructed along the northern margin of India (Chen et al., 2019), but alternative proposed locations have put it farther east along the border between India and Australia (Xu et al., 2022) or northwestern Australia (Wang et al., 2021; Zhu et al., 2012), or further west along the northeastern terminus of the East African Orogen between Arabia and India (Hu et al., 2018). The pre-rift positioning of the Lhasa Block also relates back to the timing of YZO opening, because the conventional reconstruction would place the YZO and Bangong Co-Nujiang Ocean (BNO) in parallel—making a contemporaneous opening geodynamically difficult to explain. It is crucial to better resolve the spatio-temporal context of the breakout of the Lhasa Block and the relationship between the YZO, the BNO and Neo-Tethys, as the present spectrum of proposed scenarios have significantly different implications for our understanding of the geodynamic and paleo-environmental evolution of the broader Tethyan region. Yet, direct constraints on the late Paleozoic paleogeography of the Lhasa Block are sparse.

Here we present the results of a paleomagnetic and geochronological investigation of volcanic and sedimentary rocks of the middle Permian Luobadui Formation from the Lhasa Block. We apply these new data to better constrain the paleo-position of the Lhasa Block during the middle Permian and explore the implications of its paleogeography on the broader evolution of the Neo-Tethys Ocean.

2. Geological Setting and Sampling

Upper Paleozoic strata are widely distributed in the Lhasa Block. Upper Carboniferous–lower Permian strata are characterized by glaciomarine diamictites that were then common across large areas of Gondwana, whereas middle to upper Permian strata are mainly distributed in the central part of the Lhasa Block, where they are characterized by marine sedimentary rocks, locally interbedded with volcanics (Figure 1a). The latter include an early (middle Permian) phase of basaltic volcanism and a later (middle to late Permian) phase of intermediate to silicic volcanism (Geng et al., 2009).

The most extensive exposure of the older volcanic phase, referred to as the Luobadui Formation, is located north of Luobadui village, Chundui Township, Lhunzhub County (Geng et al., 2009). The Luobadui Formation consists of a volcano-sedimentary sequence, including marine carbonates and sandstones, interspersed with volcanic layers of basalt, basaltic breccia, tuff, and andesite. The sedimentary rocks are interpreted to have been deposited in a shallow marine environment, and include fusulinids (e.g., *Sumatrina* sp., *Verbeekina* sp., *Neoschwagerina* sp.) and corals (*Iranophyllum* sp.) (CGS, 2003; Huang et al., 2022) that attest to a Wordian–Capitanian depositional age. The volcanic rocks, which have been associated with a continental-arc environment on the basis of their composition and geochemistry, have been dated to 267.5 ± 1.2 Ma by U-Pb on zircon (Wang et al., 2022).

For our paleomagnetic investigation, we collected a total of 244 oriented paleomagnetic samples from 30 sites within the Leqingla section of the Luobadui Formation (30°01.985' N, 90°58.846' E). The sampling strategy was designed to target two distinct lithological units (Figures 1b and 1c, Figure S1a in Supporting Information S1): (a) 21 volcanic sites from the middle part of the section, yielding a minimum of five samples per site (totaling 167 samples), and (b) nine limestone sites from the lower part of the section, with a minimum of five samples per site (totaling 77 samples). The middle Permian rocks form a thrust sheet over Jurassic–Cretaceous sedimentary rocks, and are overlain by Eocene volcanic rocks (CGS, 2003; Geng et al., 2009; Figure 1b). To enable a fold test, samples were collected from limestone beds with varying structural attitudes associated with faults that do not cut through the Jurassic–Cretaceous strata (CGS, 2003; Figure 1b, Figure S1a in Supporting Information S1). Bulk volcanic and sandstone samples were also collected for zircon U-Pb dating to constrain eruption ages and sedimentary provenance (Figure 1c, Figure S1a in Supporting Information S1). Additional details concerning the sampling site and sampling procedures, as well as descriptions of the rock magnetic and paleomagnetic measurement methods, are presented in the Supporting Information S1.

3. Results

Cathodoluminescence images of the volcanic zircon grains reveal most of them to be euhedral to subhedral (~50–150 μm long) with distinct oscillatory bands (Figure S1b in Supporting Information S1), suggesting they are magmatic in origin. After excluding zircon grains with less than 90% concordance, we calculated a weighted mean $^{206}Pb/^{238}U$ age of 267.8 ± 5 Ma (n = 10, MSWD = 0.072; Figure 1c, Figure S1b, Table S1 in Supporting

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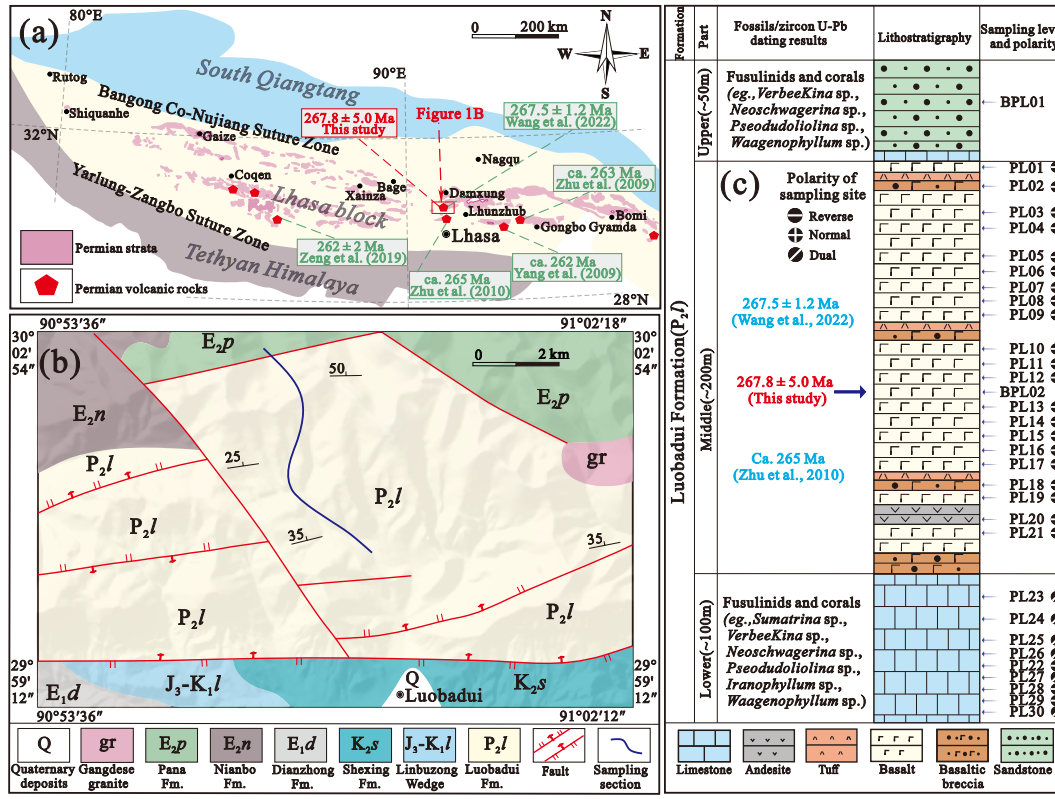


Figure 1. (a) Tectonic map of Tibetan blocks and suture zones, illustrating the study area, Permian strata distribution (modified from Wang et al., 2021; Zhang et al., 2013), and Permian volcanic rock ages in the Lhasa Block (Wang et al., 2022; Yang et al., 2009; Zeng et al., 2019; Zhu et al., 2009, 2010). (b) Geological map of the sampling area (modified from CGS, 2003). (c) Stratigraphic sequences of the Leqingla section with sampling sites (modified from Geng et al., 2009; Wang et al., 2022).

Information S1), which is consistent with the existing radiometric (Huang et al., 2022) and paleontological (Wang et al., 2022) age constraints on the Luobadui Formation described above.

Analysis of 146 zircons in the sandstone sample revealed a dominant age peak at 571 ± 10 Ma, secondary peaks at 997 ± 13 Ma and 1,161 ± 21 Ma, and several additional minor peaks in the late Archean and early Proterozoic. A comparison of these data with the detrital zircon age spectra of other geological blocks that may have been proximal to the Lhasa Block (prior to the opening of the Neo-Tethys) is shown in Figure 2. This comparison highlights a strong similarity in the detrital zircon ages of the Lhasa Block and those from the Canning Basin of northwestern Australia, particularly the age peaks at ~571 Ma and ~1,170 Ma. This observation suggests that the Lhasa Block and the Canning Basin could have once shared a common sedimentary source, indicating a possible tectonic affinity between them.

Based on a series of rock magnetic experiments (see Supporting Information S1), we identified magnetite and hematite as the dominant magnetic carriers in the volcanic rocks. Scanning electron microscopy (SEM) and energy dispersive spectroscopy (EDS) analyses of these volcanic rocks revealed primary (magmatic) iron oxide grains with euhedral to sub-euhedral morphologies that lack visible oxidation rims or surface fractures (Figures S5a–S5c in Supporting Information S1). Magnetite is also determined to be the main magnetic carrier in the limestones, where SEM analyses recognized magnetite to occur as rounded to very rounded grains (Figures S5d–S5f in Supporting Information S1). A subsidiary iron-sulfide phase, likely pyrite and/or pyrrhotite, is observed as a core within some of the iron oxide grains (Figure S5f in Supporting Information S1), and a magnetic contribution from an iron-sulfide phase is recognizable in some of the rock magnetic experimental results (see Supporting Information S1).

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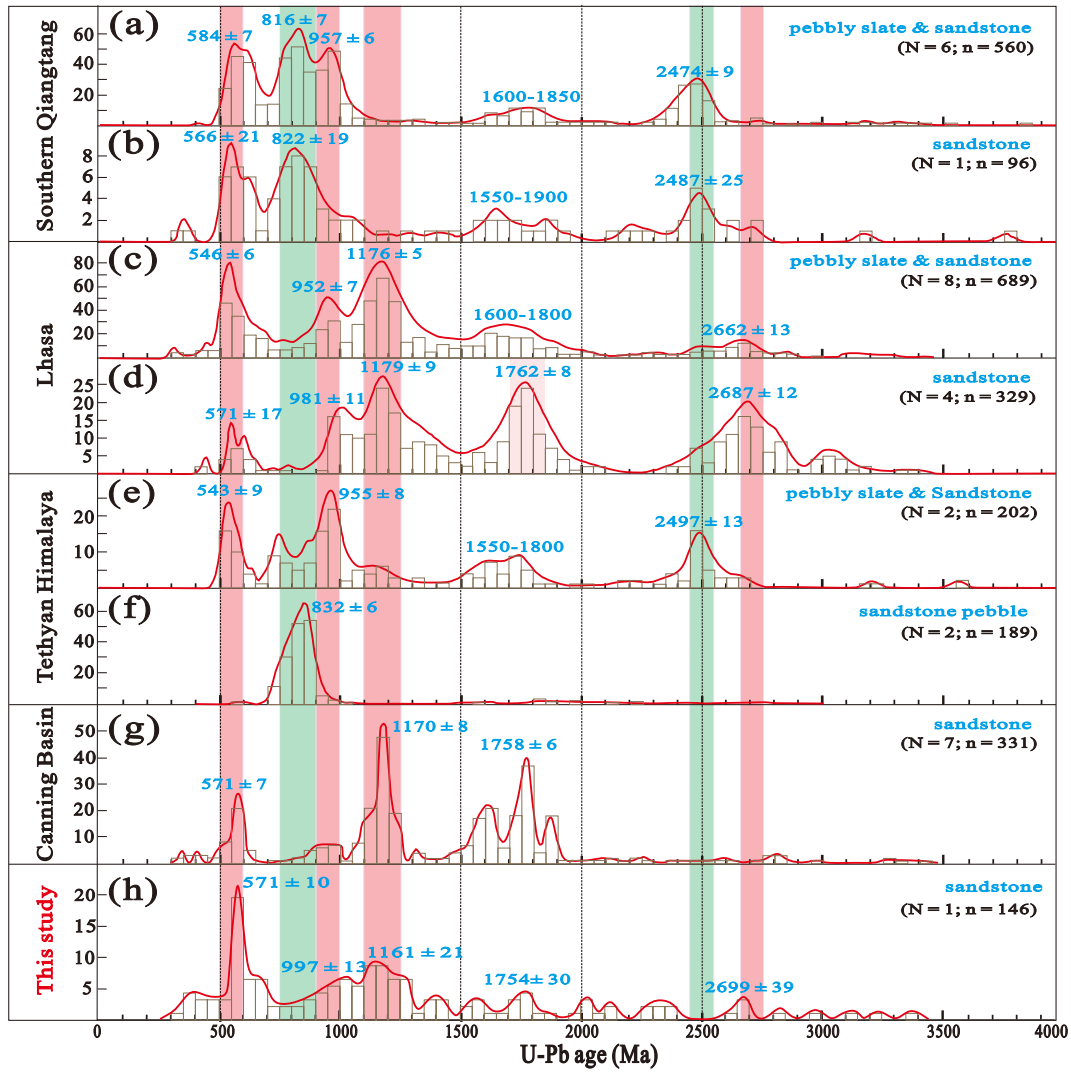


Figure 2. Detrital zircon age distributions from this study (h; data in Table S2 in Supporting Information S1) and Permo–Carboniferous strata in Tibet and the Canning Basin, Australia (a–g; modified from Wang et al., 2021). Distinctive age peaks are weighted mean ages obtained using the Isoplot 4.0 software for a given range of zircon U-Pb ages (color bands) following normal distribution. The $^{207}Pb/^{206}Pb$ ages for zircons older than 1,000 Ma and the $^{206}Pb/^{238}U$ ages for younger zircons; analyses with greater than 10% discordance were excluded. N represents the number of samples, and n denotes the total number of analyses.

Thermal demagnetization and hybrid thermal-alternating field demagnetization reveal that, with the exception of a few samples from which a stable remanent magnetization component could not be separated (Figure S10a in Supporting Information S1), most limestone and volcanic samples exhibit two distinct components of magnetization (Figures S10c–S10k in Supporting Information S1, Figure 3a–3d). A low-temperature (field) component was mostly removed by 350°C or 30 mT and has a direction, before tilt correction, very similar to that of the present-day geomagnetic field (Figure 3e). Following removal of the low-temperature (field) component, a stable high-temperature (field) component, which we designate the characteristic remanent magnetization (ChRM), decays directly toward the origin. The ChRMs of the volcanic samples were mostly demagnetized by 585°C or 75 mT, although a few maintained a stable remanence up to 680°C. These results interpreted together with the rock magnetic experiments indicate that the remanence in the volcanic samples is carried by magnetite and hematite, while in the limestone samples it is carried by magnetite. Notably, the ChRMs from both lithologies, despite differences in their magnetic carriers and the demagnetization techniques applied, are directionally consistent. Together, their ChRMs pass the fold and reversal tests, and their directional dispersion is consistent

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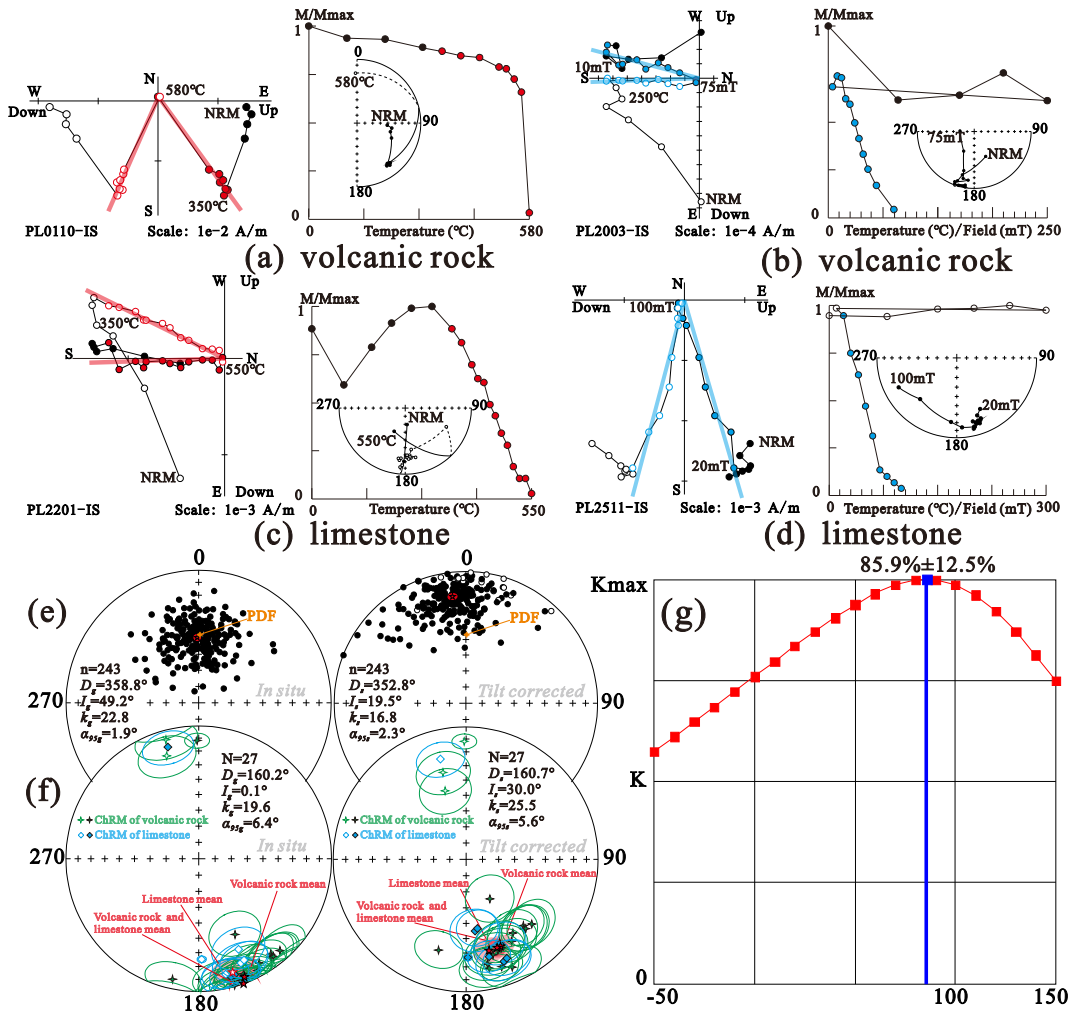


Figure 3. (a, b) Demagnetization results of volcanic rock samples (c, d) limestone samples, both in geographic coordinates. (e) Equal-area projection of low-temperature/low-field components (sample level). (f) Equal-area projection of ChRM directions (site level). (g) Progressive unfolding of ChRM directions (Watson & Enkin, 1993). Filled/open circles represent horizontal/vertical components in orthogonal plots and lower/upper hemisphere projections in stereographic plots. Red/blue circles indicate thermal/alternating field demagnetization data used in principal component analysis. Five-pointed stars mark overall mean directions with α_{95} confidence limits. NRM, natural remanent magnetization; M/Mmax, magnetization normalized to maximum magnetization; PDF, present-day geomagnetic field; n/N, number of samples/sites; D_g/D_s, I_g/I_s, declination/inclination in geographic/stratigraphic coordinates; k, Fisher precision parameter; α_{95}, radius of the 95% confidence cone.

with a population that has averaged paleosecular variation (Figures 3f and 3g, Figure S12, Table S3 in Supporting Information S1). These data yield a new paleomagnetic pole ($P_{\text{lat.}}$ = 39.6°N, P_{long} = 295.5°E, N = 27 sites, A_{95} = 5.3°) that reflects a corresponding paleolatitude of 16.7 ± 5.3°S. Additional details regarding the rock magnetic results and directional analyses are available in the Supporting Information S1.

4. Nature of the Magnetization

Our paleomagnetic analysis of the Luobadui Formation volcanic rocks and limestones identified a well-defined dual-polarity magnetization (Figure 3), which appears to have averaged out paleosecular variation. Positive fold tests further constrain the magnetization to be of pre-Jurassic–Cretaceous age. Demagnetization and rock magnetic analyses demonstrate that magnetite and hematite serve as the dominant magnetic carriers in the volcanic rocks, and SEM-EDS analyses have observed primary (magmatic) iron oxides in these rocks (Figures S5a–S5c in Supporting Information S1). In contrast, the limestones display a more intricate magnetic mineralogy, primarily

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dominated by magnetite (Figures S5d–S5f in Supporting Information S1) but with contributions from pyrrhotite, which may occur in the core of grains otherwise comprised by magnetite (Figure S5f in Supporting Information S1). This heterogeneous magnetic assemblage introduces challenges in deciphering the nature of the remanence of the limestones, necessitating careful consideration of the possibility of remagnetization.

Remagnetization is a common challenge in paleomagnetic studies of ancient carbonates from the Tibetan Plateau, often obscuring primary remanence signals (Yao et al., 2024). Nevertheless, several lines of evidence suggest the magnetization held by the limestones may represent a middle Permian remanence. SEM and EDS analyses have identified iron oxides with rounded morphologies which could represent detrital grains (Figures S5d and S5e in Supporting Information S1), and the bulk hysteresis properties of the limestones reflect an SD-PSD grain population that lacks the classic "fingerprints" of remagnetized carbonates (Figure S7 in Supporting Information S1; Jackson & Swanson-Hysell, 2012). On the other hand, a clear expression of the Verwey transition (Figure S9 in Supporting Information S1) confirms the presence of stoichiometric magnetite, which would be more consistent with an authigenic than detrital origin (if not biogenic). Despite this conflicting evidence concerning the origin(s) of the magnetite in the limestones, the fact that the ChRM directions of the volcanic rocks and the potentially remagnetized limestones are indistinct suggests that the remanence in the limestones was acquired in the middle Permian, regardless of the origin of the magnetite carrying it. One way to potentially explain this could be via the oxidation of early diagenetic iron sulfides (including pyrrhotite) driven by nearby volcanic activity, which could have transformed much of the diagenetic pyrite/pyrrhotite into magnetite. The grains we observed with iron sulfide cores enveloped by magnetite could thus represent instances where this oxidation reaction did not run to completion. In this scenario, authigenic magnetite is the primary remanence carrier, but the remanence was acquired in the middle Permian during regional volcanism. Another possibility is that detrital magnetite served as the original remanence carrier in the limestone and that localized reduction transformed some of these primary iron oxides into iron sulfides, but did not fully erase the stable remanence. In summary, although a remagnetization of the Luobadui Formation limestones cannot be entirely excluded, the available evidence suggests that they have maintained a middle Permian remanence. These data may thus shed new light on the paleo-position of the Lhasa Block and the evolution of the Neo-Tethys Ocean during the middle Permian.

5. Tectonic Implications

We integrated our newly acquired paleomagnetic data with existing late Paleozoic–early Mesozoic paleomagnetic data from the Lhasa Block and other Tethyan blocks (Table S4 in Supporting Information S1). Due to the limited availability of high-quality pre-Jurassic data for the Lhasa Block, our analysis primarily relied on recent studies. Time versus paleolatitude curves (Figure 4a) reveal that the Lhasa Block was likely part of the Cimmerian continent (including Sibumasu, South Qiangtang and Iran blocks) and rifted from Gondwana by the middle Permian. This conclusion is supported by evidence of rifting-related magmatism (300–280 Ma) and sedimentary changes along Gondwana's northern margin, which reflect that a tectonic reorganization was already underway at that time. These processes impacted multiple regions, including the Lhasa, South Qiangtang, Baoshan, Himalaya, and Panjal blocks (e.g., Dan et al., 2021; Fan et al., 2023).

The classical Tethyan kinematic model suggests that the Lhasa Block, as part of the Cimmerian continent, drifted northward after separating from Gondwana, eventually colliding with Laurasia (Cheng et al., 2023; Wei et al., 2022, 2023; Şengör et al., 1988). However, Zhu et al. (2012) suggested that transform faults along Gondwana's northern margin led to differential motions among the Cimmerian blocks. This interpretation has particular significance for the Lhasa Block, which may have been located near the eastern terminus of the Cimmerian continent, along the northwest margin of Australia, as indicated by detrital zircon provenance associations. Indeed, middle Permian foraminiferal assemblages from the Lhasa Block suggest it was isolated from both Gondwana and other blocks of the Cimmerian continent at this time (Zhang et al., 2019). Zhou et al. (2016) further proposed, on the basis of Early–Middle and Late Triassic paleomagnetic data, that the Lhasa Block did not experience rapid northward drift during the Triassic (as observed in other Cimmerian terranes), but remained in a prolonged stagnation phase. These findings suggest that the Lhasa Block may have experienced slow drift along the northwestern margin of Australia, perhaps because it was decoupled from the kinematics of neighboring Cimmerian blocks by transform faults (Figures 4b and 4c). Clearly, such "hesitant" kinematics are not unique. A similar prolonged low-latitude stasis is observed in the paleomagnetic data from Indochina, North Qiangtang, and South China blocks, which likely followed comparable trajectories after rifting from Gondwana (Ding

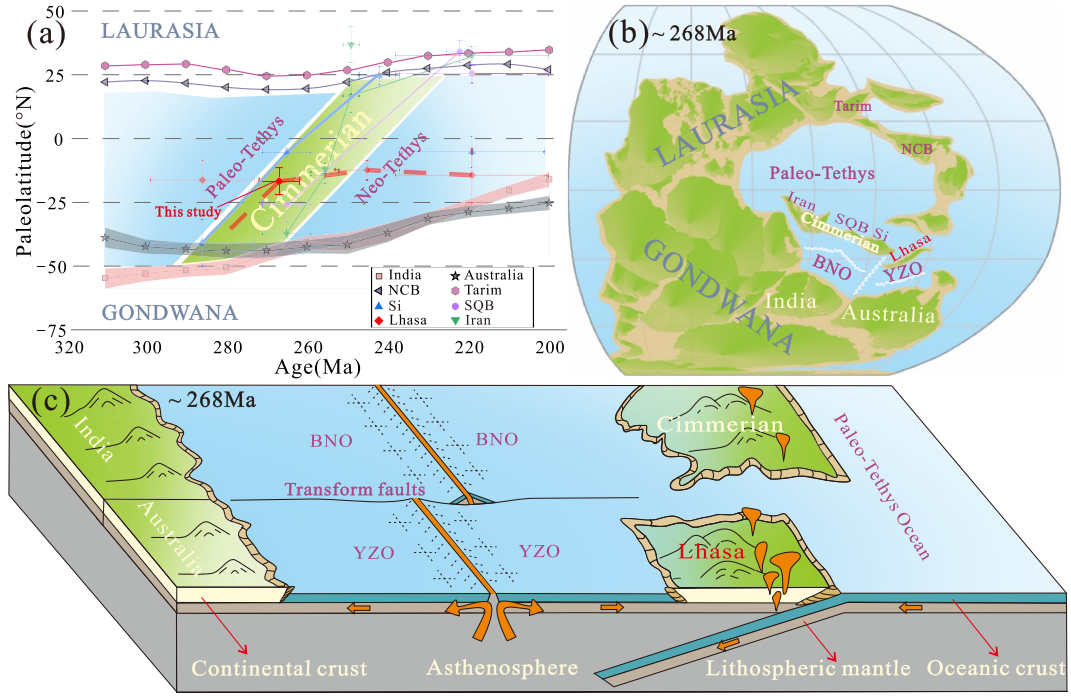


Figure 4. (a) Paleolatitude versus time plot generated from paleomagnetic data from the Lhasa Block, the Cimmerian continent (including Sibumasu, South Qiangtang and Iran blocks), Gondwana's northern margin (including the Australian and Indian sectors) Laurasia's southern margin (including North China and Tarim blocks) during 200–310 Ma. All data presented are summarized in Table S4 in Supporting Information S1; (b, c) Paleogeographic reconstructions (modified from Huang et al., 2018; Wei et al., 2022, 2023) and schematic models (modified from Li et al., 2023; G. Yang, 2022; Zhu et al., 2012) of the Lhasa Block and its adjacent tectonic blocks/plates at ~268 Ma. Abbreviation: SQB, South Qiangtang Block; Si, Sibumasu Block; NCB, North China Block. Note that our reconstruction is simplified: Argoland, which may have been located along the northwestern margin of Australia (Advokaat & van Hinsbergen, 2024), and the Tengchong Block, which may have been located near the Lhasa Block (Shen et al., 2024), are not shown due to the lack of sufficient paleomagnetic constraints.

et al., 2017; Yan et al., 2024). Therefore, based on this evidence, we hypothesize that the Lhasa Block may have been "left behind" by the main Cimmerian wave during the late Paleozoic–early Mesozoic.

Within the framework of the Wilson Cycle, plate drift is driven by oceanic spreading and subduction. Middle Permian reconstructions of the Lhasa and South Qiangtang blocks generally depict the BNO and YZO as east-west-elongated branches of the Neo-Tethys Ocean, consistent with the models of Stampfli and Borel (2002) and Ferrari et al. (2008), which integrate dynamic plate boundaries, plate buoyancy, ocean spreading rates, and major tectonic and magmatic events. The opening of these oceans is often assumed to have resulted from tectonic processes operating along Gondwana's northern margin, though the mechanisms remain debated. The rifting and northward drift of the Lhasa Block from the northern margin of Gondwana was specifically associated with the opening and growth of the YZO. Concerning this early development of the YZO, the Luobadui Formation volcanic rocks are especially important in indicating the presence of an active continental arc along the Lhasa Block during the middle Permian. However, the polarity of this arc is unknown, and both southward subduction (Geng et al., 2009) and northward subduction (Wang et al., 2022) have been proposed. The widespread occurrence of ophiolites (Liu et al., 2020) and subduction-related volcanism (Yan et al., 2023) in the Lhasa Block further complicates this interpretation. We propose a sequential model involving back-arc spreading associated with the southward subduction of the Paleo-Tethys beneath the northern margin of the Lhasa Block, following the opening of the YZO. This process likely caused the Lhasa Block to migrate rapidly northward at a rate of about 30 cm/yr. Northward subduction of the YZO began beneath the southern margin of the Lhasa Block in the middle Permian. The rollback of this subduction zone could have been counterbalanced by spreading of the YZO, which would have kept the Lhasa Block at approximately the same paleolatitude, significantly reducing its drift rate, until subduction ceased or the YZO spreading ridge was subducted (Figures 4b and 4c).

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Although we contend that this proposed scenario best explains the available observations, several critical issues remain unresolved. Specifically: (a) We cannot completely exclude the possibility that the new paleomagnetic data reported herein are not a pre-Jurassic–Cretaceous remagnetization. (b) The existing Permian and Triassic paleomagnetic data for the Lhasa Block, primarily obtained from sandstones (Dong et al., 1991; Ran et al., 2012) and limestones (Zhou et al., 2016), have been challenged due to their sparsity and uncertainties regarding the reliability and origin of their magnetizations (Yao et al., 2024). (c) The paleolatitude variation curve (Figure 4a) indicates that the Lhasa Block was situated at ∼16°S from at least the middle Permian to the Late Triassic. However, during this same interval, significant northward paleolatitudinal shifts were witnessed in both Gondwana and Laurasia. Furthermore, Late Triassic paleomagnetic constraints from the Lhasa Block would seem to place it near the margin of Gondwana again. A plausible explanation is thus that the Lhasa Block underwent a large-scale remagnetization during the Triassic, prior to rifting from Gondwana.

Despite these limitations, which hinder a conclusive reconstruction of the kinematic history of the Lhasa Block from the late Paleozoic to early Mesozoic, we argue that the paleomagnetic data reported here are most likely of middle Permian age and thus provide valuable insights into the separation of the Lhasa Block from Gondwana and the initial stages of Neo-Tethys Ocean opening. To better resolve the outstanding uncertainties regarding the post-rift history of the Lhasa Block and its potentially prolonged paleolatitudinal stagnation, there is an urgent need for additional high-quality Permian and Triassic paleomagnetic data from this block.

6. Conclusions

We have presented a paleomagnetic pole ($P_{lat.}$ = 39.6°N, P_{long} = 295.5°E, N = 27 sites, A_{95} = 5.3°) which we interpret to represent a middle Permian (267.8 ± 5.0 Ma) remanence, which was obtained from limestones and volcanic rocks of the Lhasa Block. These new paleomagnetic data place the Lhasa Block at 16.7 ± 5.3°S at that time. By further integrating new detrital zircon geochronology data and additional geological evidence, we propose that the Lhasa Block rifted from the Australian sector of northern Gondwana before the middle Permian, with the Yarlung-Zangbo Ocean and the Bangong Co-Nujiang Ocean, as branches of the Neo-Tethys, likely coexisting in an east-west configuration.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Data used in this study (Table S1–S5 in Supporting Information S1) can be accessed at Zenodo (Xing, 2025).

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